PAOLA ARGUELLO PASCUALLI

Vancouver, BC | paolaap@student.ubc.ca | github.com/paola-arguello

Computational genomics researcher with training in statistical genetics, epigenomics and functional genomics. Interested in brain gene regulation and the shared genetic architecture of psychiatric disorders.

EDUCATION

MSc, Medical Genetics, University of British Columbia 2025

Supervisors: Dr. Jessica Dennis (Genetic Epidemiology) and Dr. Michael Kobor (Social Epigenetics)

Project: Assessing the role of genetic admixture on genetic and epigenetic health biomarkers

- Quality-controlled and imputed genotype array data; inferred global and local ancestry.
- Modelled regional population structure and historical migrations in Costa Rica (first-author manuscript, below).
- Processed DNA methylation array data and computed polygenic scores and epigenetic clocks to test how admixture affects these biomarkers.

BSc (Honours), Genomic Sciences, National Autonomous University of Mexico (UNAM) 2019

Coursework included systems biology, gene regulatory networks, dynamical modelling and machine learning.

RESEARCH EXPERIENCE

Graduate Research Assistant, Elliott Lab, Simon Fraser University Winter 2023

Canadian COVID-19 host genetics cohort (HostSeq)

- Computed polygenic risk scores and tested for ancestry-specific variation in severity associations. Second author, PLoS Genetics 2024.

Visiting Research Scholar, Rehkopf Lab, Stanford University Jan – Mar 2023

Integrating environmental and cultural perspectives in genetic research (MSc, Friedman Award)

- Applied Python-based migration models to estimate the timing and magnitude of admixture events, alongside environmental and cultural context.

Visiting Research Scholar, Moreno Lab, LANGEBIO, Mexico May – Dec 2022

Integrating environmental and cultural perspectives in genetic research (MSc, Friedman Award)

- Developed Python-based migration models estimating the timing and magnitude of population admixture events.

Research Assistant, Gräff Neuroepigenetics Lab, EPFL, Switzerland Oct 2018 – 2019

Unraveling molecular signatures of epigenetic priming in memory formation

- Analyzed RNA-seq and ChIP-seq data on the effect of HDAC inhibitors on epigenetic priming in memory formation.
- Performed mouse work (certified in Laboratory Animal Science, University of Lausanne).

Research Intern, Torres-Padilla Epigenetics Lab, Helmholtz Zentrum München, Germany Winter 2018

- Analyzed public ChIP-seq data validating epigenetic marks of totipotency. Co-author, Nature Genetics 2022.

PUBLICATIONS

Arguello-Pascualli P, et al. Genome-wide ancestry inference reveals regional structure and historical migrations in Costa Rica. *In preparation for submission to PLoS Genetics.*

Bruxel EM, Rovaris DL, … **Arguello-Pascualli PY**, … Montalvo-Ortiz JL (2025). Psychiatric genetics in the diverse landscape of Latin American populations. *Nature Genetics*, 57(5), 1074–1088.

Garg E, **Arguello-Pascualli P**, … Elliott LT (2024). Canadian COVID-19 host genetics cohort replicates known severity associations. *PLoS Genetics*, 20(3), e1011192.

Nakatani T, Lin J, Ji F, … **Arguello-Pascualli P**, … Torres-Padilla ME (2022). DNA replication fork speed underlies cell fate changes and promotes reprogramming. *Nature Genetics*, 54(3), 318–327.

RESEARCH SOFTWARE

LIRA: Likelihood Ratio Meta-Analysis web application, UBC 2025

- Designed and built a web application for likelihood-ratio meta-analysis, from requirements to deployment on a secure server (HTML, CSS, Bitbucket), with developer documentation and user guides.

HONOURS AND AWARDS

Four Year Fellowship (4YF), UBC | $18,200 CAD 2021 – 2025

Li Tze Fong Fellowship (competitive), UBC Affiliated Fellowships | $25,000 CAD 2021 – 2022

Friedman Award for Scholars in Health (competitive) | $38,000 CAD 2020

Funded research internships at LANGEBIO, Mexico (2022) and Stanford University (2023).

TEACHING

Teaching Assistant, Genome Informatics (BMEG 400E), UBC Jan – Apr 2021, 2022

- Created course content, tutorials and assignments on bioinformatic tools (github.com/BMEGGenoInfo/Assignments).
- Guest lecture, "Polygenic scores as clinical tools: a long way to go," with a hands-on PGS assignment (2021, 2022).

Teaching Assistant, Genetics and Human Heredity (BIOL 344), UBC Sept – Dec 2024

- Led two 2-hour tutorials per week.

Teaching Assistant, Human Genomics and Medical Genetics (MEDG 420), UBC Sept – Dec 2022

Teaching Assistant, Clinical Genomics (MEDG 580), UBC Sept – Dec 2021

TRAINING

Hands-on Introduction to Rare Disease Genomics, BC Children's and Women's Hospital July 2021

Polygenic Scores Hands-on Workshop, STAGE Training Program in Advanced Genetic Epidemiology June 2021

TECHNICAL SKILLS

Programming: R, Bash, Python, Perl, SQL (MySQL), HTML/CSS

Statistical genetics: PLINK, genotype imputation, polygenic scores (PRSice, PRS-CS, LDpred), ancestry inference (ADMIXTURE, RFMix, PCA)

Epigenomics and sequencing: DNA methylation arrays (minfi, ewastools, wateRmelon, epigenetic clocks); RNA-seq (STAR/HISAT2, featureCounts/Salmon, DESeq2/edgeR); ChIP-seq (Bowtie2, MACS2, deepTools, HOMER)

Computing: SLURM on institutional servers, conda and containers, Git/GitHub, Snakemake/Nextflow, R Markdown and Jupyter